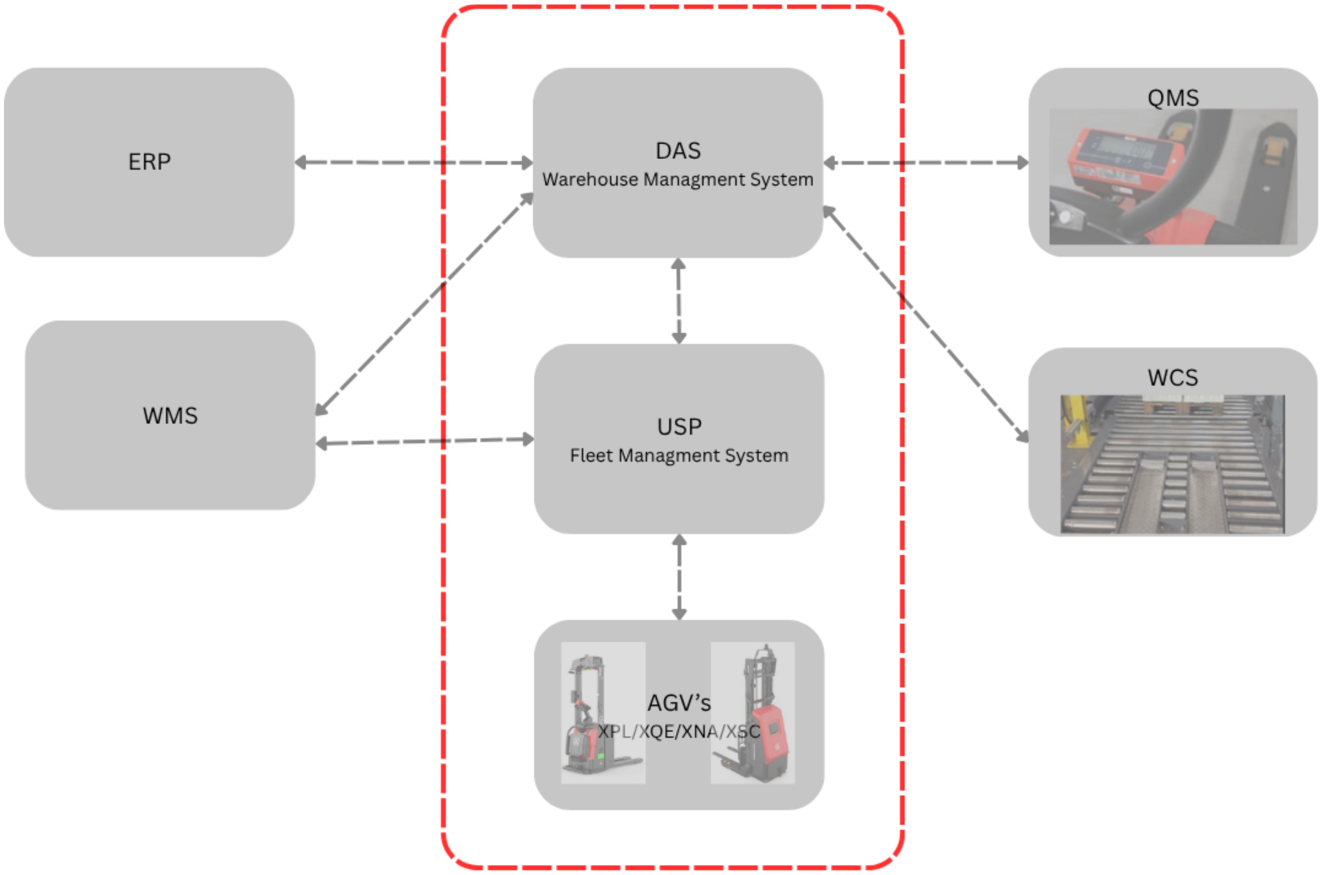
ERP
WMS
DAS
Warehouse Managment System
USP
Fleet Managment System
AGV's
XPL/XQE/XNA/XSC
QMS
WCS

USP Server (Fleet manager)

Required for / done by	USP operation, including scheduling, task handling and path-planning support.
Minimum specification	• CPU: Intel i7. • GPU: integrated GPU. • RAM: 64 GB RAM. • Storage: 2 TB SSD.
Software / operating principle	• USP recommendation references USP v2.3.1 and Ubuntu 22.04.5 LTS 64-bit as the software environment. • Use the USP server as a headless server where possible, managed remotely instead of through local monitor/browser usage. • Avoid monitoring browsers or non-essential local applications directly on the USP server.
Acceptance before commissioning	• USP server installed, powered and accessible for configuration. • Connected to the AGV/server network by wired network connection. • No unnecessary local applications left running on the server.
Comments / notes	• Consider an additional backup server. • USP requires task input from a WMS/DAS or another approved task trigger.

DAS Server (Warehouse management system)

Required for / done by	WMS operation and warehouse task layer above USP.
Minimum specification	• Processor / CPU: E-2224 quad-core CPU at 3.4 GHz. • CPU type: desktop-grade CPU. • RAM: 32 GB RAM. • Primary storage: 256 GB SSD. • Additional storage: 1 TB HDD.
Acceptance before commissioning	• Server installed, powered and available for the DAS scope. • Network connection prepared with final IP allocation, static DNS, gateway and required routing. • Wired internet/network connection prepared. • Server reserved for the AGV/DAS function unless another architecture is formally approved.
Comments / notes	• Consider an additional backup server • DAS is the next layer above USP. • DAS can connect to the customer WMS or ERP system.

Wi-Fi / WLAN Infrastructure

Required for / done by	AGV-to-server communication, USP/DAS connectivity and commissioning readiness.
Design requirements	• AP quantity and placement suitable for the full operating area, avoiding unnecessary mutual interference. • Adjacent AP channels staggered using 1, 6 and 11 where applicable. • AP installation angle: 7 to 9 degrees tilt. • AP installation location: target coverage range 15 m and installation height approx. 3 m. • Wall attenuation and physical obstructions considered in the design. • Operating frequency: 2.4 GHz / 5 GHz.
Performance targets	• Signal strength: greater than -65 dB. • Real-time bandwidth: greater than 5 Mbps. • Network delay: less than 100 ms. • Packet-loss design target: less than 4%; final route-test acceptance requires no packet loss during the test. • IP planning: machine + server allocation to include 10 IP, static DNS, gateway and number of channels as required by the project.
Comments / notes	• Any packet loss or interruption must be investigated and resolved before acceptance. • Possible causes include other wireless devices, physical obstructions, machinery or electrical equipment. • Wired connection for server. (LAN Cable) • Wi-Fi network for AGVs and add-ons such as PDA.

Wi-Fi Checking Procedure - Execution Method

Test coverage	Full AGV operating route, including travel lanes, pickup/drop-off points, intersections, charging areas and any area where AGV communication is required.
Test tool	Laptop using Terminal on MacBook or Command Prompt on Windows.
Test command	ping <actual AGV/server network IP address>. Replace the example IP with the actual IP address used by the project network.
Test interval	Repeat the test every 5 meters along the AGV route.
Acceptance criteria	• No packet loss during route testing. • Reliable Wi-Fi coverage across the full AGV operating path. • Stable response time with no communication interruptions.
Failure condition	Any packet loss, unstable response time or connection interruption is a commissioning hold point until investigated and resolved.

Step-by-Step Procedure

Step	Required action
1. Identify network IP address	• Open Network & Internet Settings. • Select Wi-Fi or Ethernet as applicable. • Under Properties, note the IPv4 address.
2. Open command interface	• On MacBook, open Terminal. • On Windows, open Command Prompt.
3. Run ping test	• Enter the ping command to the actual AGV/server network IP address. • Example: ping 192.168.100.89. • Replace the example IP with the real project network IP address.
4. Review results	• Check packet loss. • Check response-time stability. • Check for interruptions or timeouts.
5. Investigate issues	• Review possible interference from other wireless devices. • Check obstructions, walls, racking, machinery or electrical equipment. • Document corrective action before acceptance.
6. Test full route	Conduct testing along the complete AGV travel and operation path.
7. Repeat every 5 m	Repeat testing every 5 meters to confirm coverage and communication reliability throughout the route.

Evidence to Record

Required evidence	- Route/location tested every 5 m. - IP address used for the ping test. - Packet-loss result. - Response-time stability / interruptions. - Corrective action and retest result if a failure is found.
Final acceptance statement	Wi-Fi/WLAN is accepted only when testing confirms no packet loss and reliable communication coverage across the full AGV operating path.

```
C:\Users\ArnabMitra>ping 192.168.100.89

Pinging 192.168.100.89 with 32 bytes of data:
Reply from 192.168.100.89: bytes=32 time=236ms TTL=128
Reply from 192.168.100.89: bytes=32 time=3ms TTL=128
Reply from 192.168.100.89: bytes=32 time=5ms TTL=128
Reply from 192.168.100.89: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.100.89:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 236ms, Average = 61ms
```

Network Ports and Protocols

Required for / done by	USP scheduling, AGV communication, charging-pile communication where applicable, RCS/label integration where applicable, and server-to-device connectivity.
Commissioning rule	Firewall/VLAN/routing rules must allow the listed ports and protocols. If any required port is blocked or untested, commissioning remains on HOLD for the affected function.
Evidence required	Port availability test, service reachability test, successful message exchange or web/service connection test, and confirmation of network path between server(s), AGVs, chargers and add-ons.

USP Fleet Manager Ports and Protocols

Service	Port / Protocol	Scope	Purpose	Required action / notes
nginx (HTTP web server)	80 HTTP / WebSocket	External	Scheduling Management Page	Allow access between approved client/service network and USP server. Confirm the scheduling management page is reachable before commissioning.
EMQX (MQTT)	1883 TCP	External	Vehicle connection scheduling	Allow AGV-to-USP MQTT communication. Confirm AGV can connect and exchange state/order messages with USP.
Charging Pile Service	8087 UDP	External	Charging-pile status/reporting to scheduling service	Required when charging-pile integration is in scope. The charging pile uses port 503 as the client and reports status to port 8087 of the scheduling service server.

XPL Fleet manager Ports and Protocols

Service	Port / Protocol	Scope	Purpose	Required action / notes
HTTP / JSON	8103 HTTP / WebSocket	External	Scheduling Management Page / RCS JSON communication	Allow RCS communication to the relevant server/service. Confirm HTTP/WebSocket reachability and JSON payload exchange before commissioning.

Payload and Interface Validation

Interface	Direction / message type	Content handled	Commissioning validation
USP - AGV	USP to AGV MQTT order payload	Order/task instructions including order ID, task ID, nodes, edges, map ID, positions and action commands such as fork-control actions.	Send or simulate a USP order and confirm the AGV receives the task/path-planning instruction.
AGV - USP	AGV to USP state payload	AGV status includes position, map ID, battery state, operating mode, safety state, errors, fork status and task/order identifiers.	Confirm USP receives live AGV state updates without delay, disconnects or message errors.
RCS / Label interface	HTTP/JSON label-info payload	Label and task synchronization information including bind ID, AMR IP/port, sync area, order ID, task ID, node ID and label/QR code data.	Confirm RCS JSON message is received and parsed by the relevant receiving service.

Suitcase Setup - POC / Mini Demo Showroom

Purpose	Recommended compact setup for proof of concept, training or small showroom demonstration.
Typical scope	• AGV + manual charger. • Server with 32 GB RAM. • Portable Wi-Fi router with range up to approx. 50-75 m, depending on site conditions.
Provided setup	A configured machine and server are provided together with the portable router.
Operator steps	• Follow the machine implementation manual. • Complete scanning. • Add maps and routes. • Showcase or perform tasks using Fleet Manager on the server or on the machine display.